

IMPULSE RESPONSE IN SECOND ORDER, LINEAR ODE WITH CONSTANT COEFFICIENTS

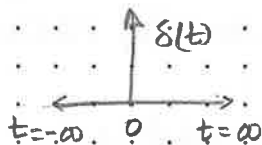
IMPULSE RESPONSE $[g(t)]$

$$A g'' + B g' + C g = \delta(t)$$

initial state.

$$g(0) = 0$$

$$g'(0) = 0$$



Letting
 $A=1$

$$g'' + B g' + C g = \delta(t)$$

Recasting the ODE into a homogenous ODE with appropriate initial states

$$g'' + B g' + C g = 0$$

$$g(0) = 0$$

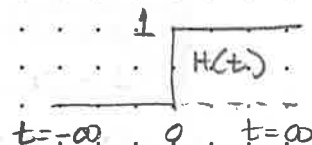
$$g'(0) = 1$$

- ① Key idea: Solve a null/homogenous ODE with the new initial states.
 $g(0) = 0$; $g'(0) = 1$

- ② Key idea: I am looking for null solutions starting with $g(0) = 0$ & initial slope $g'(0) = 1$

JUMP RESPONSE $[r(t)]$

$$A r'' + B r' + C r = CH(t)$$



$$r'' + B r' + C r = CH(t)$$

with initial states

$$r(0) = r'(0) = 0$$

Particular solution (r_p):

$$H(t) = 1 \quad t \geq 0$$

We know this!

What if $r_p = 1$?

$$r_p'' = 0, \quad r_p' = 0$$

$$\Rightarrow \cancel{r_p''} + B \cancel{r_p'} + C \cdot \cancel{1} = CH(t)$$

$$\Rightarrow C \cdot 1 = C \quad \boxed{r_p = 1}$$

One can now find out the two null solutions, & follow through with the general solution.

IMPULSE RESPONSE (CONTINUED)

Solution:

$$\begin{cases} q'' + \beta q' + Cq = 0 \\ q(0) = 0; q'(0) = 1 \end{cases}$$

characteristic equation:

$$s^2 + \beta s + C = 0$$

$\Rightarrow$ We get s_1, s_2
for the null solution

General (null) solution:

$$q(t) = C_1 e^{s_1 t} + C_2 e^{s_2 t}$$

Apply initial states:

$$q(0) = 0 \Rightarrow C_1 + C_2 = 0$$

$$\Rightarrow \boxed{C_2 = -C_1}$$

$$q'(t) = C_1 s_1 e^{s_1 t} + C_2 s_2 e^{s_2 t}$$

$$q'(0) = C_1 s_1 + C_2 s_2 = 1$$

$$\Rightarrow C_1 s_1 - C_1 s_2 = 1$$

$$\Rightarrow \boxed{C_1 = \frac{1}{s_1 - s_2}} \quad \boxed{C_2 = \frac{-1}{s_1 - s_2}}$$

$$q(t) = \frac{e^{s_1 t} - e^{s_2 t}}{s_1 - s_2}$$

$\hookrightarrow$ Fundamental solution
from which you can create
all solutions.

$\hookrightarrow$ This tells us about the
growth rate in 2nd order
ODE

$\hookrightarrow$ Analogous to e^{at} , growth
rate in 1st order ODEs

JUMP RESPONSE (CONTINUED)

However, I will
explore an alternative
approach in this
section.

$\rightarrow$ Since I have the impulse response

[How is $r(t)$ connected to $q(t)$]

$\Rightarrow$ How is the step function
connected to δ -function

$$[\text{Jump function}] = [\text{Integral of } \delta\text{-function}]$$

$$[\text{Jump response}] = [\text{Integral of } \delta\text{-response}]$$

⑤ Set up the δ -response $q(t)$
as an integral:

$$r(t) = \int_0^t \left(\frac{e^{s_1 \tau} - e^{s_2 \tau}}{s_1 - s_2} \right) d\tau$$

$\hookrightarrow$ $q(\tau)$ with
a dummy
variable τ .

IMPULSE RESPONSE (Continued)

$$q'' + Bq' + Cq = 0$$

① $B=0$ No Damping.

$$s^2 + C = 0$$

$$\Rightarrow s = \pm iC \text{ or } \pm i\omega_n$$

Where $\omega_n \rightarrow$ Natural frequency

$$q(t) = \frac{e^{i\omega_n t} - e^{-i\omega_n t}}{i\omega_n - (-i\omega_n)}$$

$$\Rightarrow q(t) = \frac{e^{i\omega_n t} - e^{-i\omega_n t}}{2i\omega_n}$$

$$\Rightarrow q(t) = \frac{\sin(\omega_n t)}{\omega_n}$$

② Under-damping $B^2 - 4C < 0$

$$s = \frac{-B \pm \sqrt{B^2 - 4C}}{2}$$

$$\Rightarrow s = \left(-\frac{B}{2}\right) \pm i\omega_d$$

$$q(t) = \frac{e^{(-B/2 + i\omega_d)t} - e^{(-B/2 - i\omega_d)t}}{(-B/2 + i\omega_d) - (-B/2 - i\omega_d)}$$

$$q(t) = \frac{(e^{-Bt/2}) \sin(\omega_d t)}{\omega_d}$$

③ Critical Damping $B^2 - 4C = 0$

$$(s + \frac{B}{2})^2 = 0 \Rightarrow s = -\frac{B}{2} = -\frac{B}{2}$$

$$q(t) = t e^{-Bt/2}$$

JUMP RESPONSE (CONTINUED)

II. Evaluate the integral.

$$r(t) = \left(\frac{1}{s_1 - s_2}\right) \left[\frac{e^{s_1 t}}{s_1} - \frac{e^{s_2 t}}{s_2} \right]_0^t$$

III. Apply limits of integration, 0 to t.

$$r(t) = \left(\frac{1}{s_1 - s_2}\right) \left[\left(\frac{e^{s_1 t}}{s_1} - \frac{e^{s_2 t}}{s_2}\right) - \left(\frac{1}{s_1} - \frac{1}{s_2}\right) \right]$$

IV. Simplify the term:

$$\left(\frac{1}{s_1} - \frac{1}{s_2}\right) = \frac{s_2 - s_1}{s_1 s_2}$$

$$r(t) = \left(\frac{1}{s_1 - s_2}\right) \left[\frac{e^{s_1 t}}{s_1} - \frac{e^{s_2 t}}{s_2} - \frac{s_2 - s_1}{s_1 s_2} \right]$$

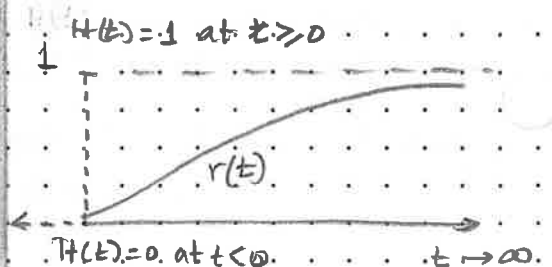
V. Distribute & write in standard form:

$$r(t) = \frac{s_2 e^{s_1 t}}{s_1 s_2 (s_1 - s_2)} - \frac{s_1 e^{s_2 t}}{s_1 s_2 (s_1 - s_2)} + \frac{1}{s_1 s_2}$$

$$\Rightarrow r(t) = \left(\frac{1}{s_1 s_2}\right) \left[1 - \frac{s_1 e^{s_1 t} - s_2 e^{s_2 t}}{s_1 - s_2} \right]$$

hence step or jump response

{ where $s_1 \neq s_2$ }



JUMP RESPONSE (CONTINUED)

Overdamped case:

$s_1 \neq s_2$ Real & distinct roots

$$r(t) = \left(\frac{1}{s_1 s_2} \right) \left(1 - \frac{s_1 e^{s_2 t} - s_2 e^{s_1 t}}{s_1 - s_2} \right)$$

critically damped case:

$s_1 = s_2 = -\omega_n$ Real, repeated root

Apply L'Hôpital Rule as $s_2 \rightarrow s_1$:

$$r(t) = \lim_{s_2 \rightarrow s_1} \left(\frac{1}{s_1 s_2} \right) \left(1 - \frac{s_1 e^{s_2 t} - s_2 e^{s_1 t}}{s_1 - s_2} \right)$$

Evaluating the limit & substituting $s_1 = -\omega_n$:

$$r(t) = \left(\frac{1}{\omega_n^2} \right) [1 - (1 + \omega_n t) e^{-\omega_n t}]$$

Underdamped case:

$$s_1 + s_2 = -\zeta \omega_n + i \omega_d$$

Complex
conjugate
roots

$$\text{where } \left[\zeta = \frac{\beta}{2/\omega_n} \right]$$

from $A r'' + B r' + C r = CH(t)$

$$\& [\omega_d = \sqrt{1 - \zeta^2}]$$

Using Euler's formula $e^{i\theta} = \cos\theta + i \sin\theta$

$$r(t) = \left(\frac{1}{\omega_n^2} \right) \left[1 - e^{-\zeta \omega_n t} \left(\cos(\omega_d t) + \frac{\zeta}{\sqrt{1 - \zeta^2}} \sin \omega_d t \right) \right]$$

OR

$$r(t) = \left(\frac{1}{\omega_n^2} \right) \left[1 - \frac{1}{\sqrt{1 - \zeta^2}} e^{-\zeta \omega_n t} \cos(\omega_d t - \phi) \right]$$

{ where $\phi = \tan^{-1} \left(\frac{\zeta}{\sqrt{1 - \zeta^2}} \right)$ }